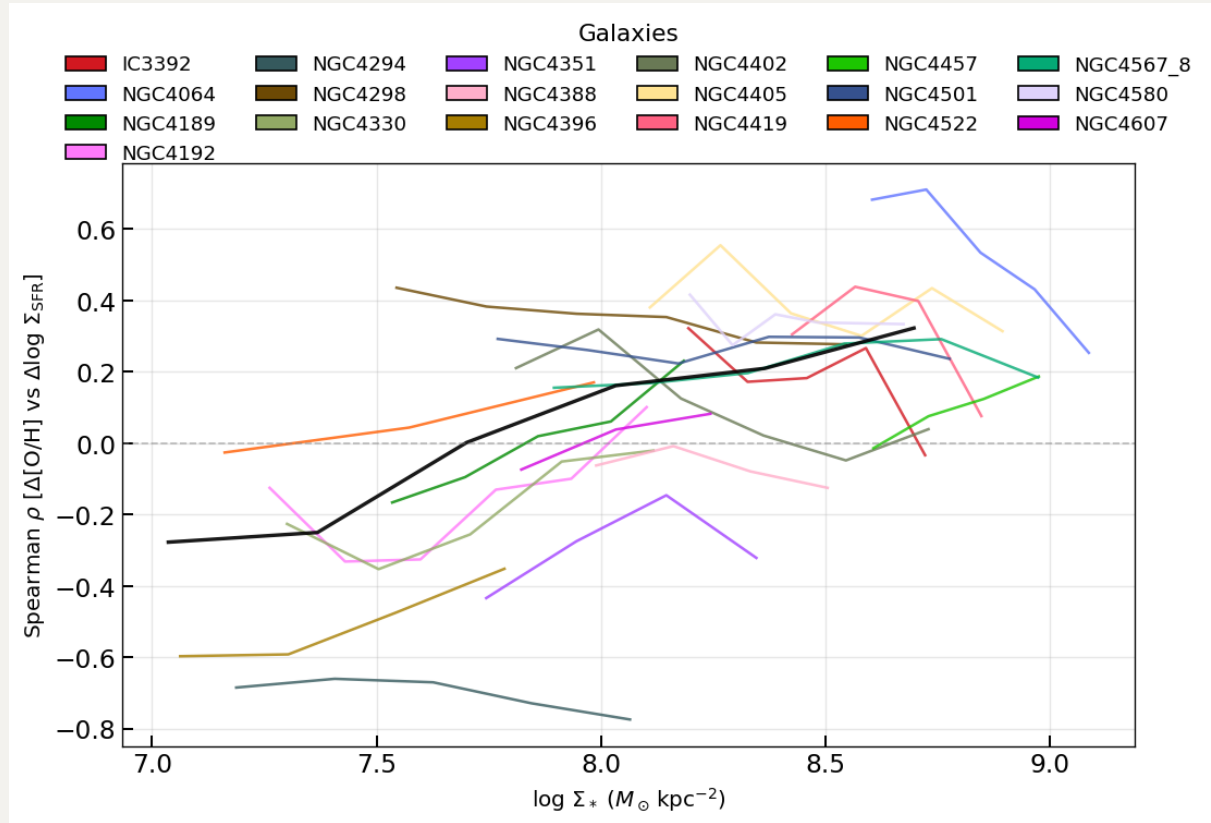
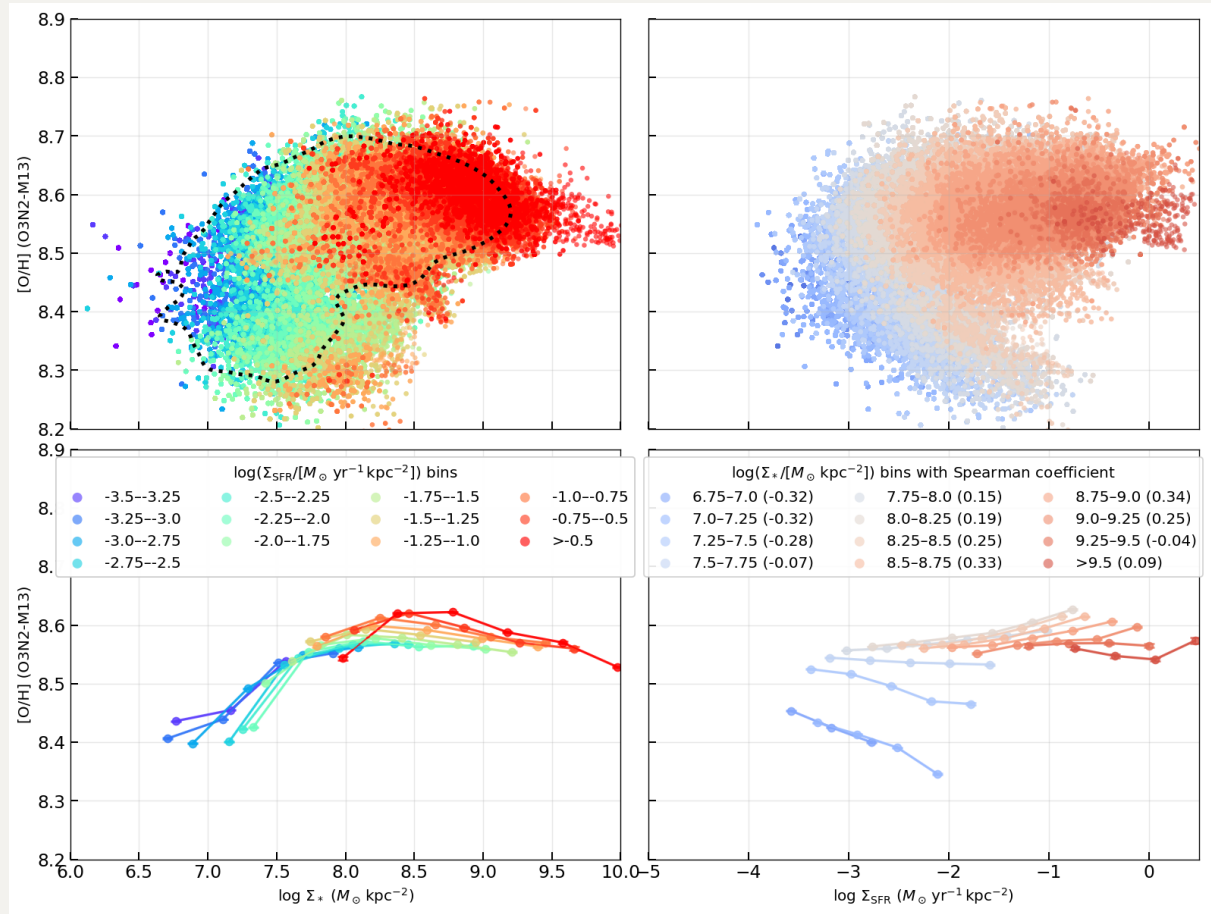


20260602 Turnover Galaxies in OH SFR Space



Since we have more statistics, now I set the bin size at 0.25 dex so that we have more median trends. Here, we can clearly see that in [O/H]- Σ_{SFR} parameter space, these galaxies form two branches: upward and downward. The goal is to find if there exist a galaxy that occupy both branches, which is the turnover galaxy.



Below i show loop this plot for 26 galaxies, but we can pay attention to the following:

- Maybe turnover galaxies: NGC4189, NGC4192, NGC4522 (very weak)

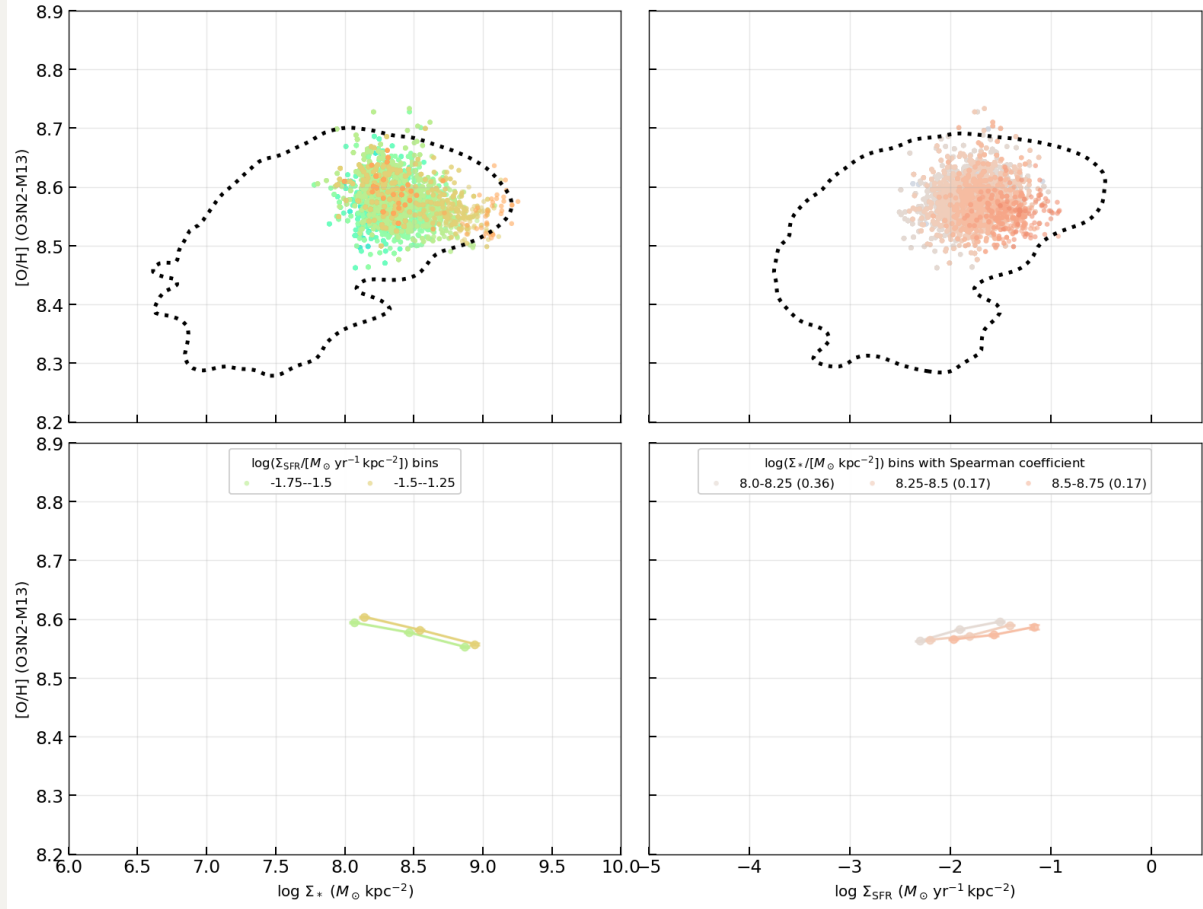
Sadly, after go through the O/H-SFR space one-by-one, i can still only find 4189 and 4192 as the potential signal of turnover galaxies. Ideally, we need a galaxy that clearly span stellar mass in HII regions at least from 7.0-9.0 with enough statistics so that we may see strong signal of turnover.

I also list some galaxies with clear positive or negative signals:

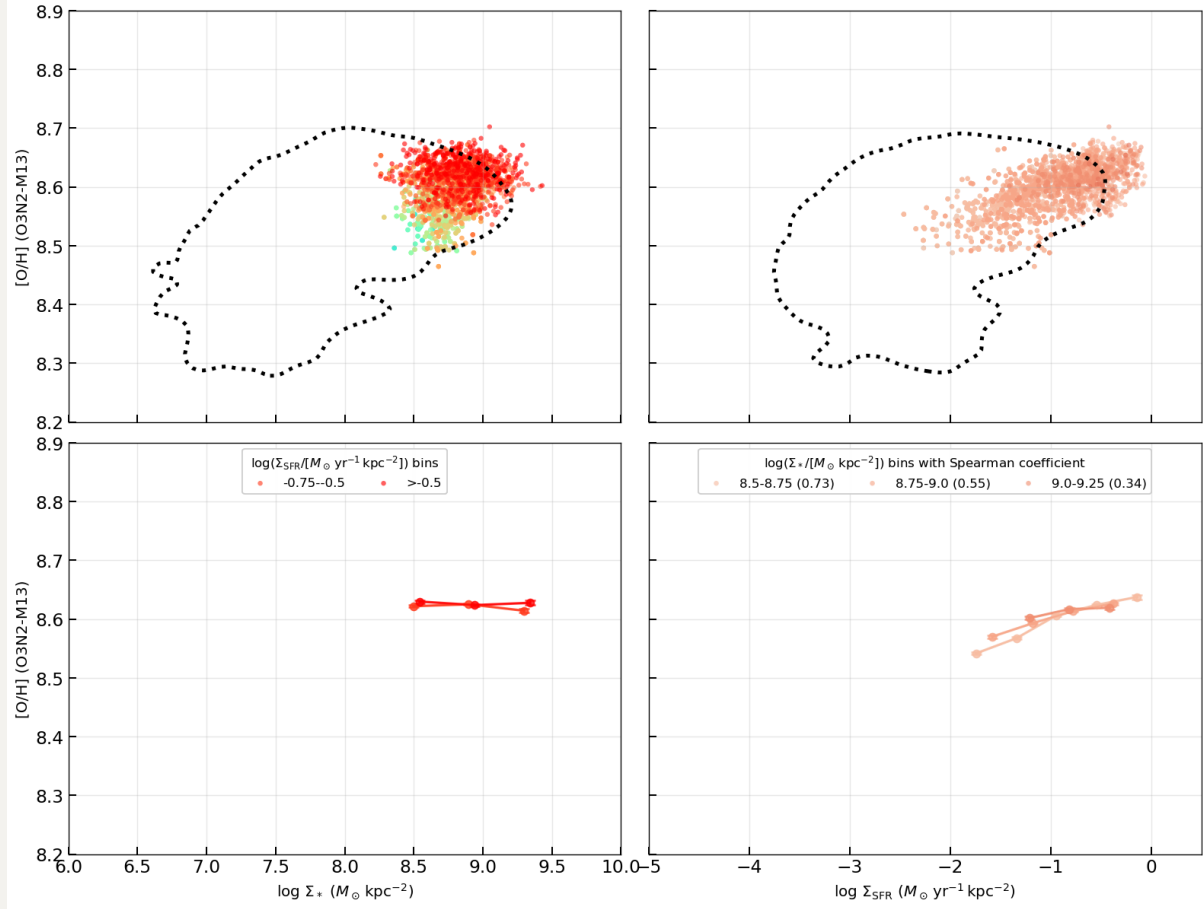
- Strong upward galaxies: NGC4064, NGC4405, NGC4501, NGC4580, NGC4606, NGC4694
- Strong downward galaxies: NGC4294, NGC4396

NGC4294 seems to be another metallicity outlier that we need to be careful.

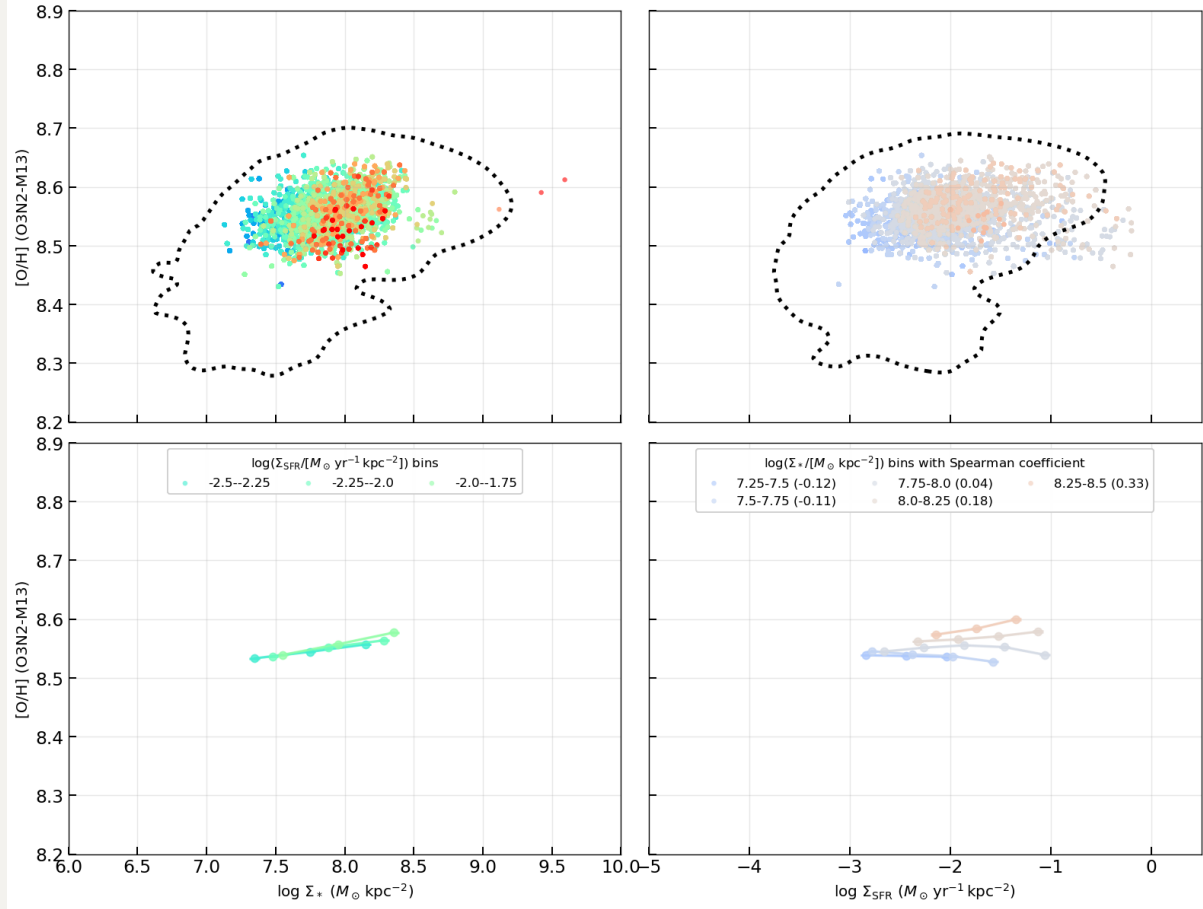
IC3392: rMZR and Z_gas-Sigma_SFR relation (further_ready)



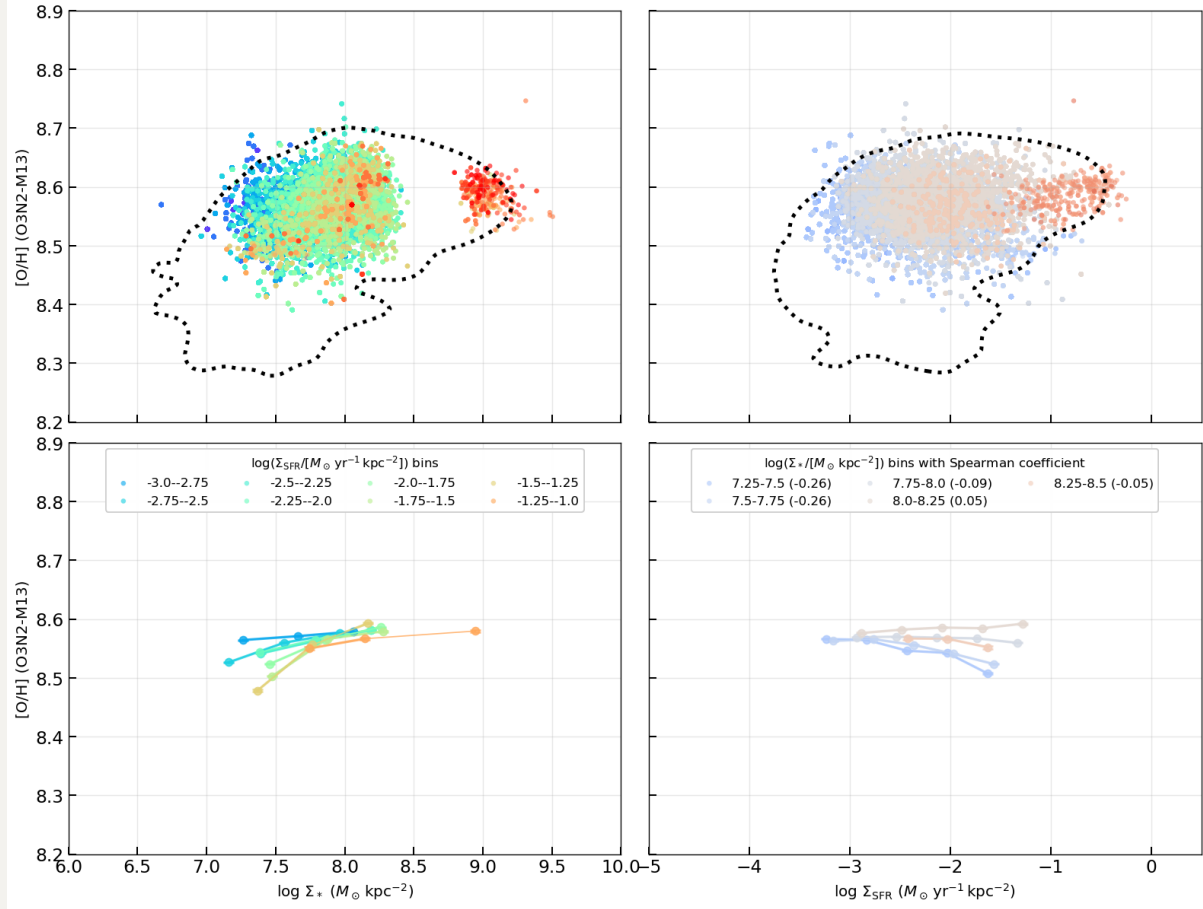
NGC4064: rMZR and Z_gas-Sigma_SFR relation (further_ready)



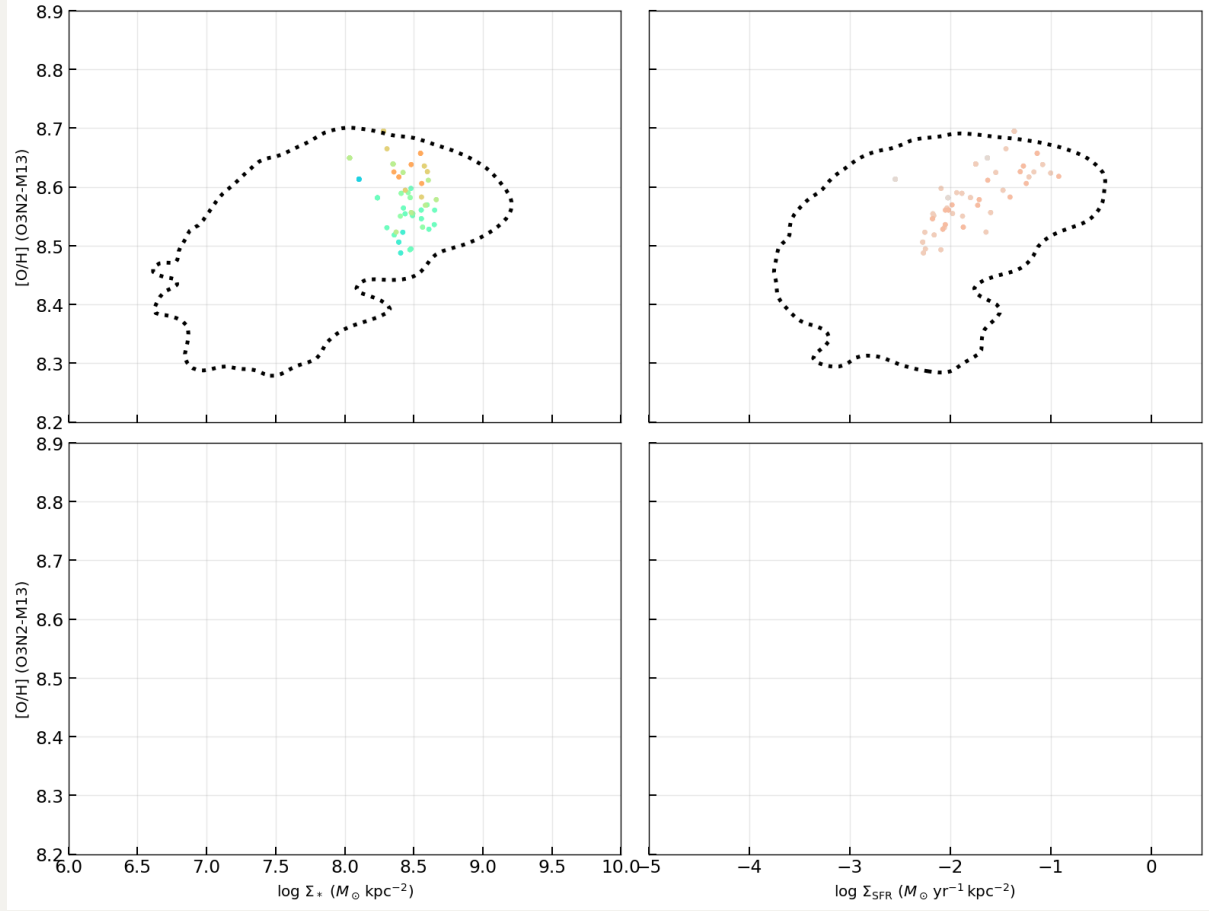
NGC4189: rMZR and Z_gas-Sigma_SFR relation (further_ready)



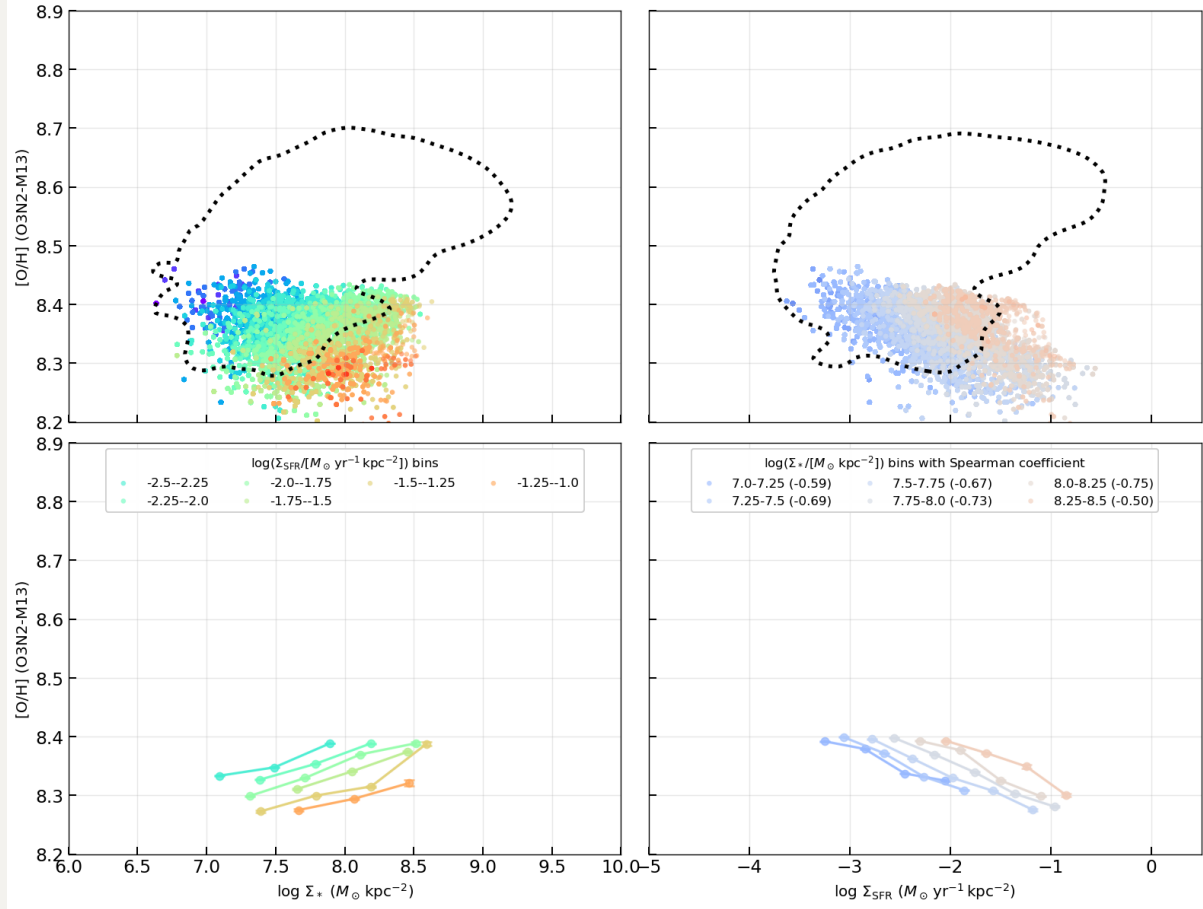
NGC4192: rMZR and Z_{gas}-Sigma_{SFR} relation (further_ready)



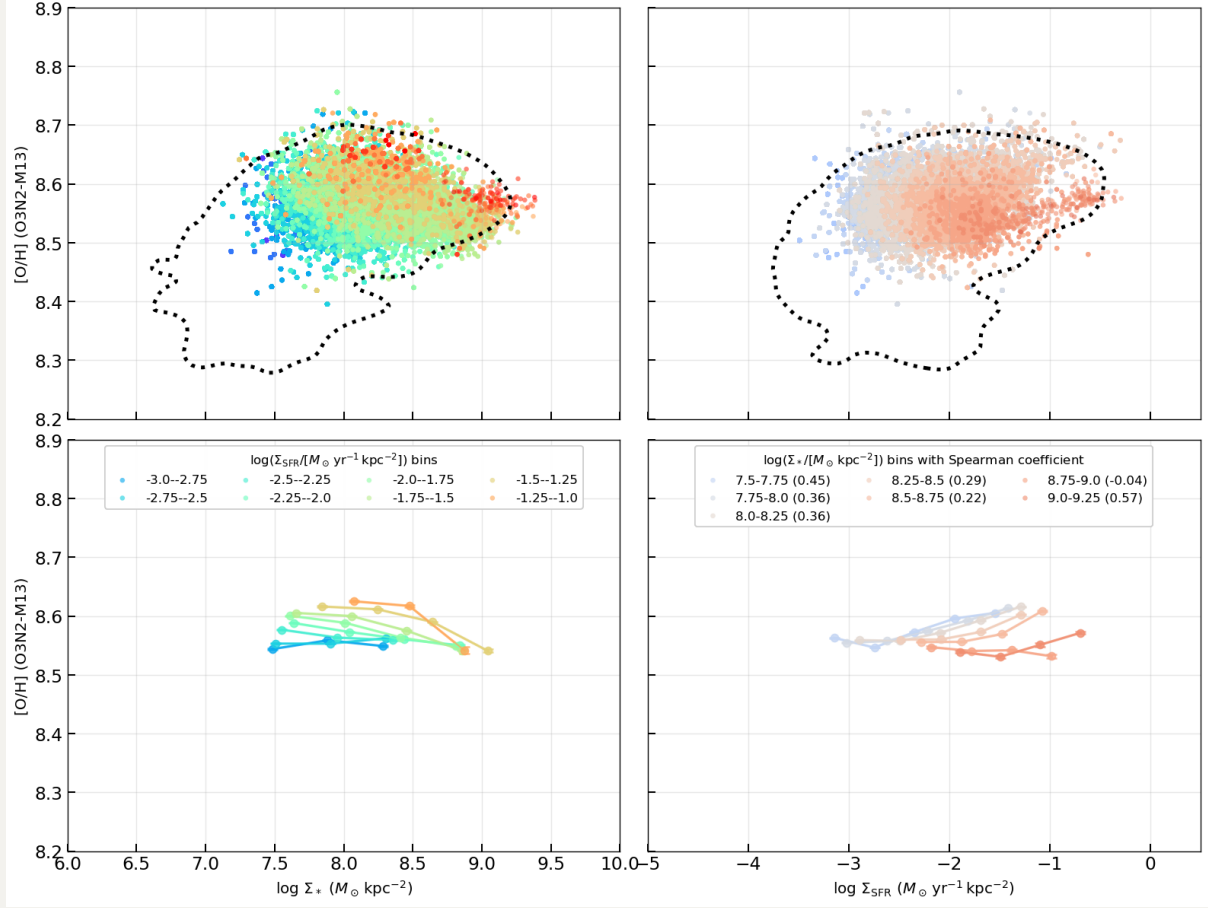
NGC4293: rMZR and Z_gas-Sigma_SFR relation (further_ready)



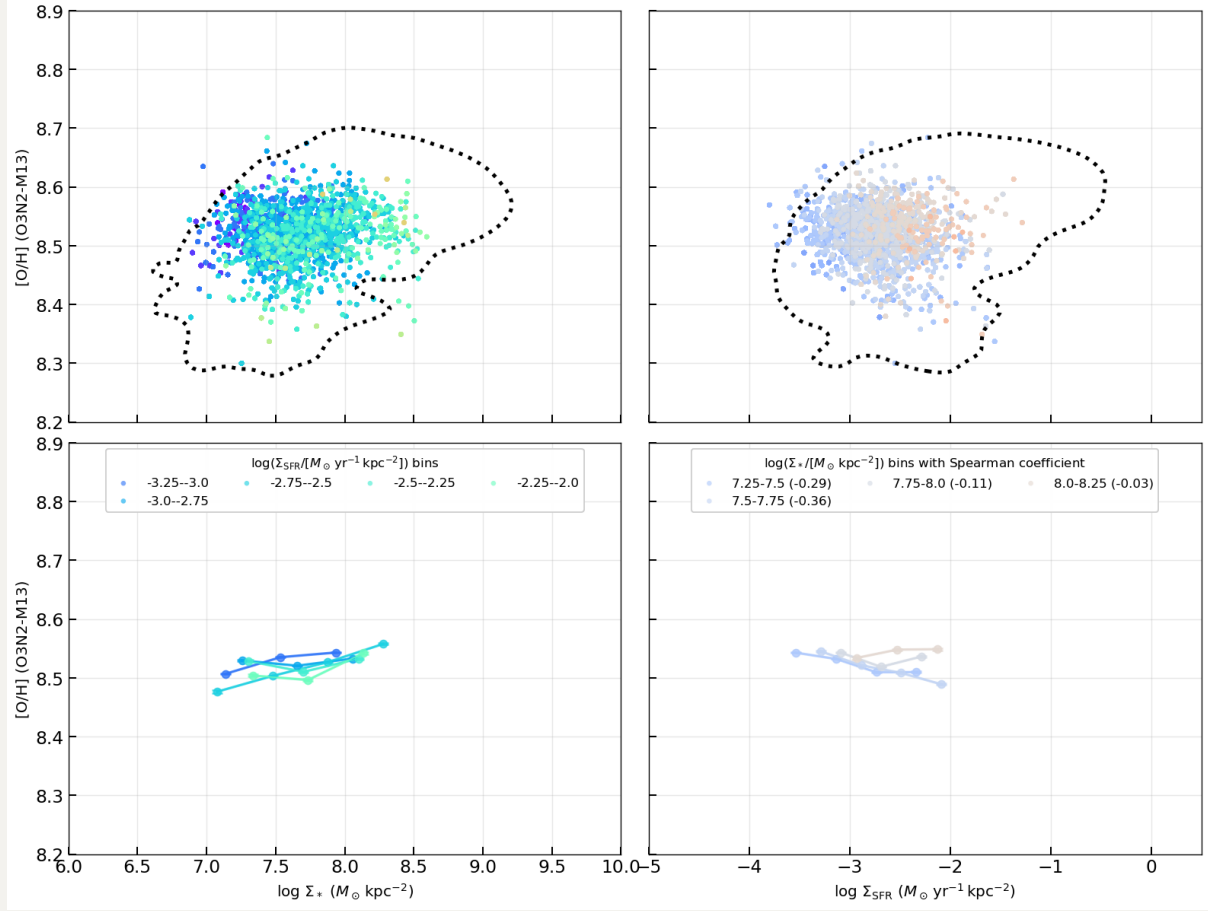
NGC4294: rMZR and Z_gas-Sigma_SFR relation (further_ready)



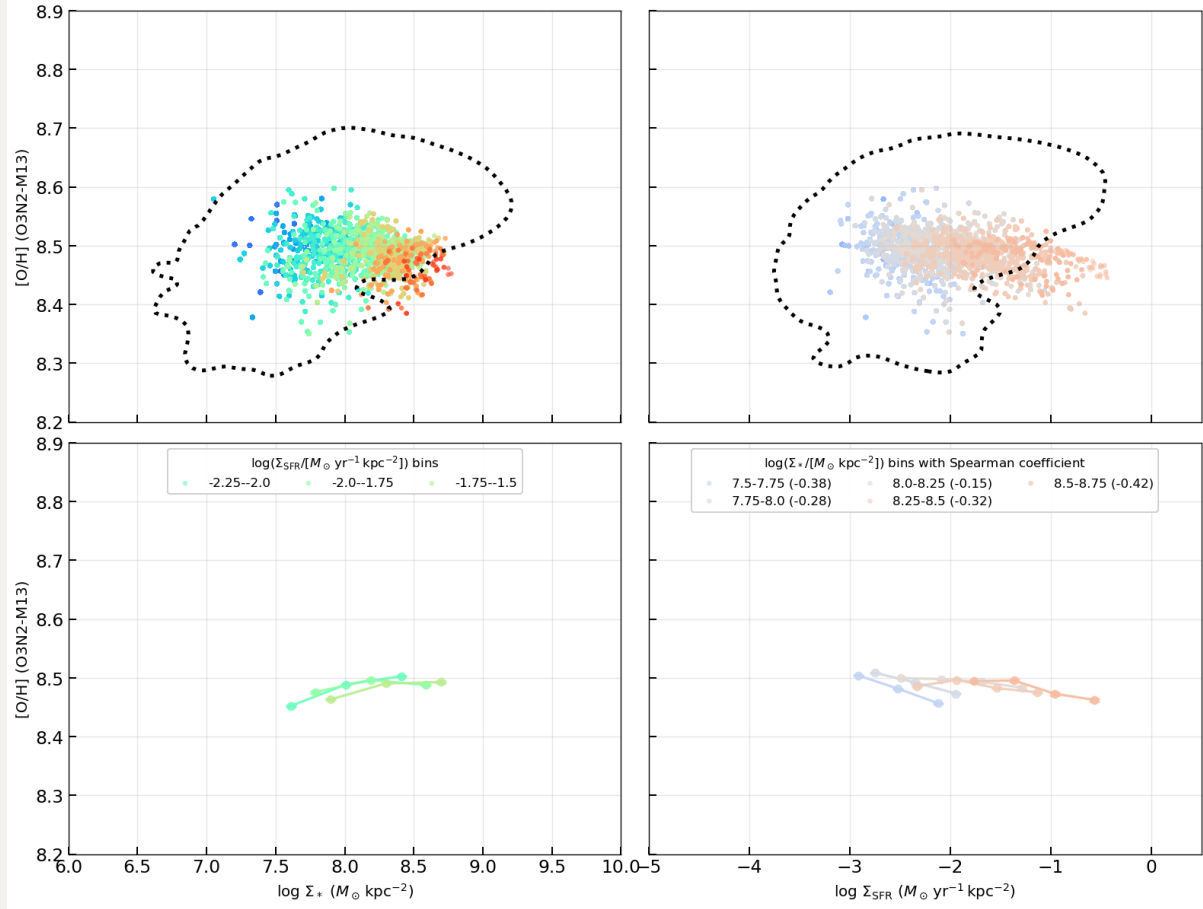
NGC4298: rMZR and Z_{gas}-Sigma_{SFR} relation (extended_fake)



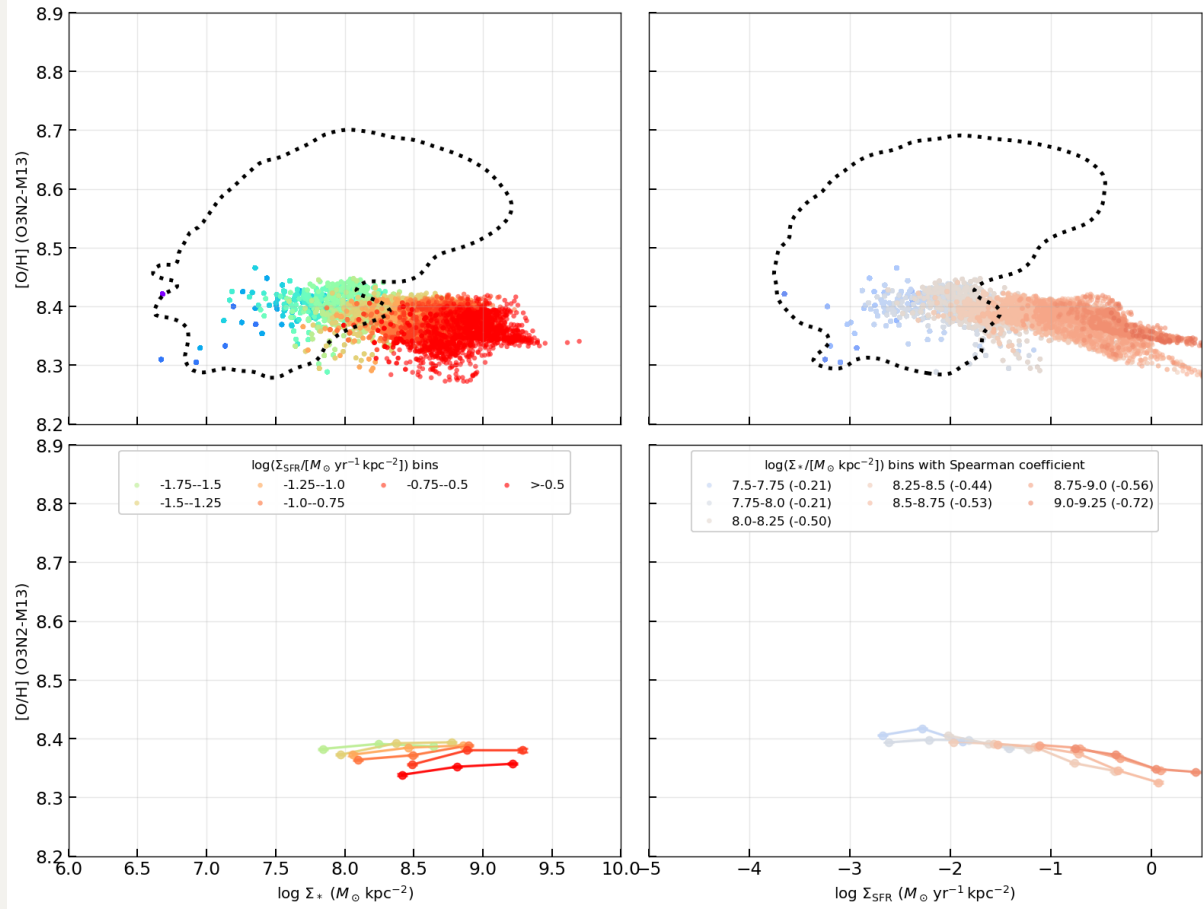
NGC4330: rMZR and Z_{gas}-Sigma_{SFR} relation (extended_fake)



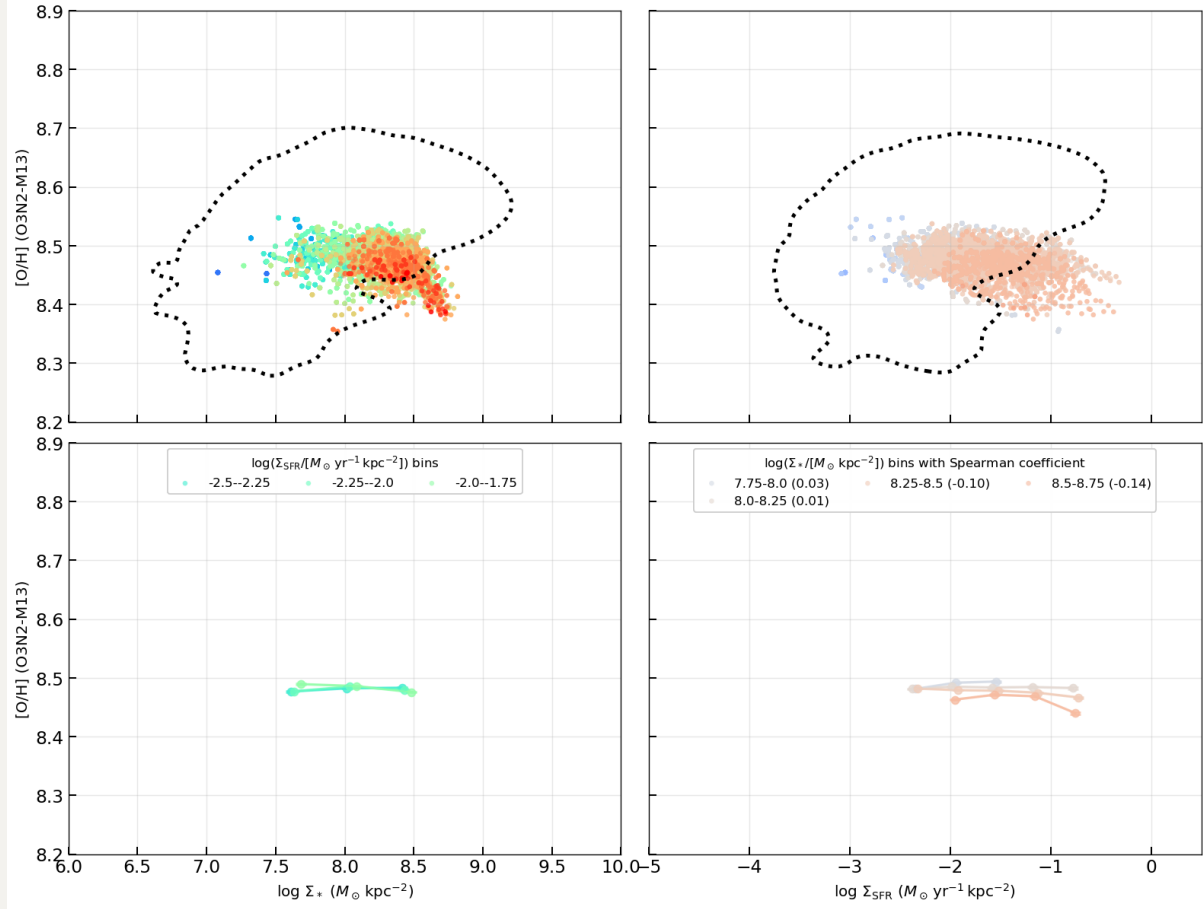
NGC4351: rMZR and Z_gas-Sigma_SFR relation (further_ready)



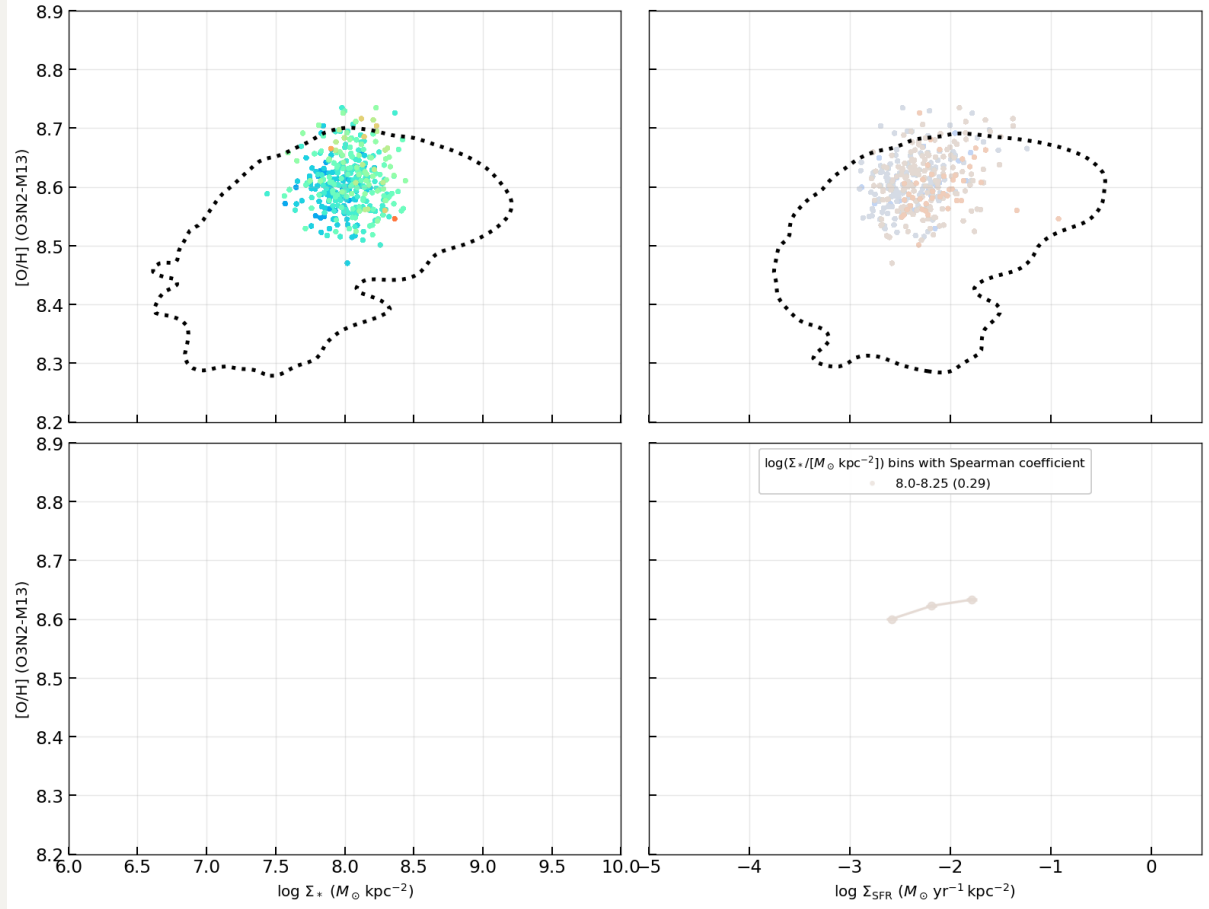
NGC4383: rMZR and Z_gas-Sigma_SFR relation (further_ready)



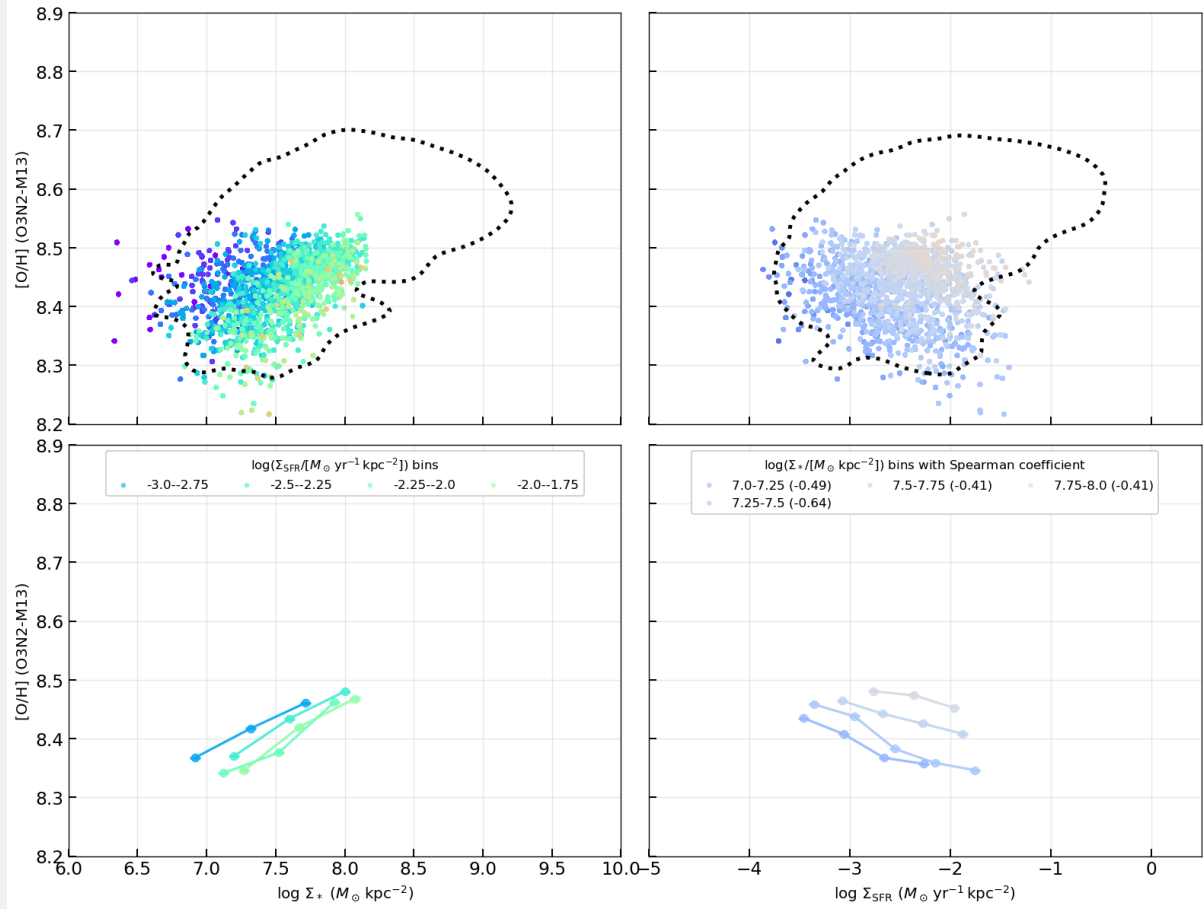
NGC4388: rMZR and Z_gas-Sigma_SFR relation (further_ready)



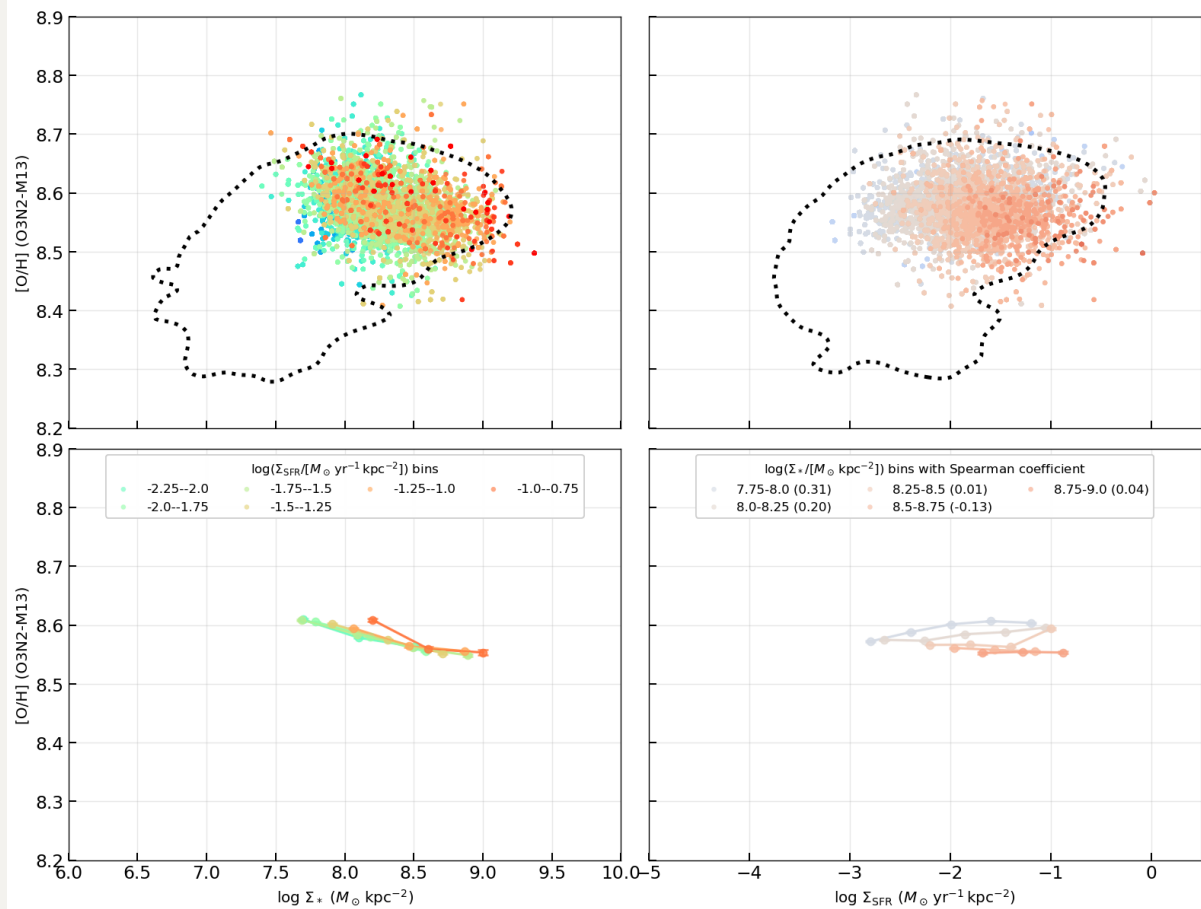
NGC4394: rMZR and Z_gas-Sigma_SFR relation (further_ready)



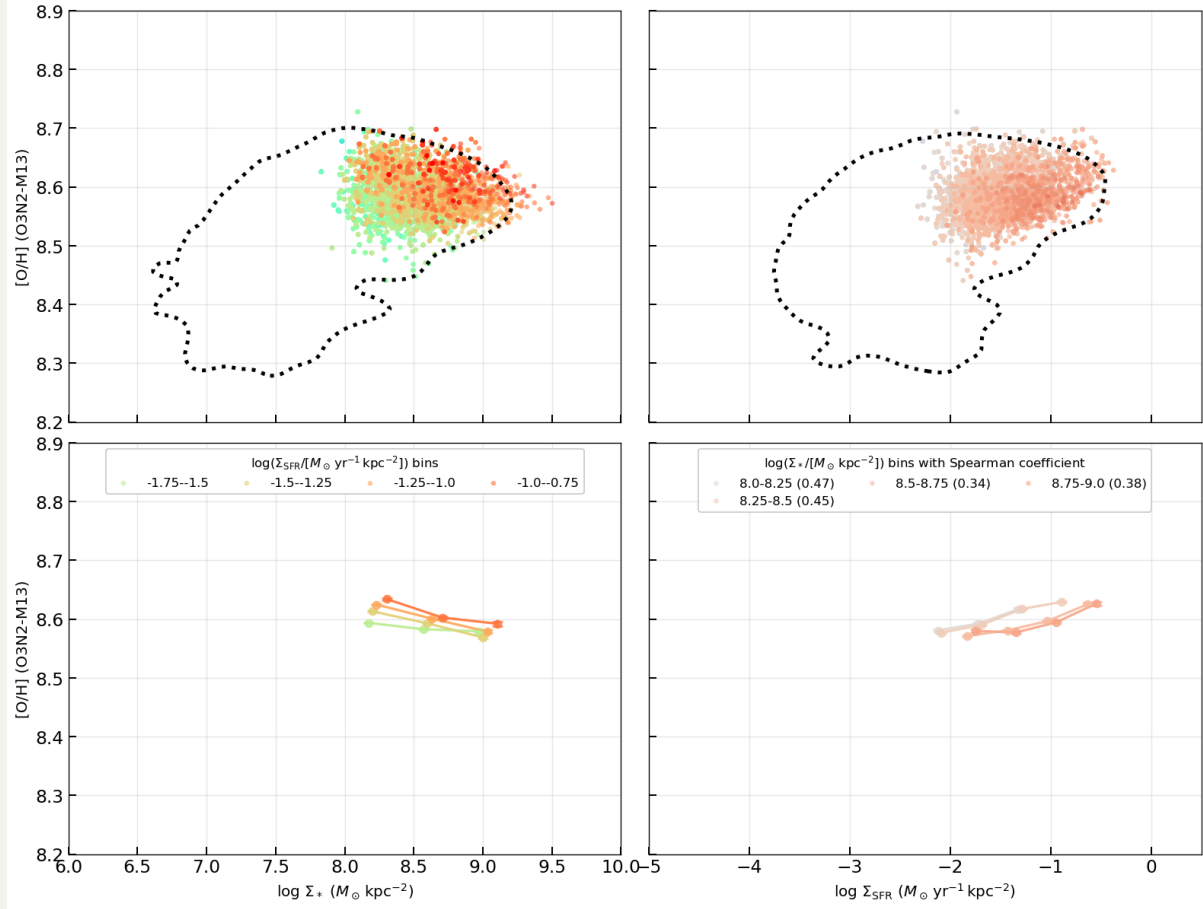
NGC4396: rMZR and Z_gas-Sigma_SFR relation (further_ready)



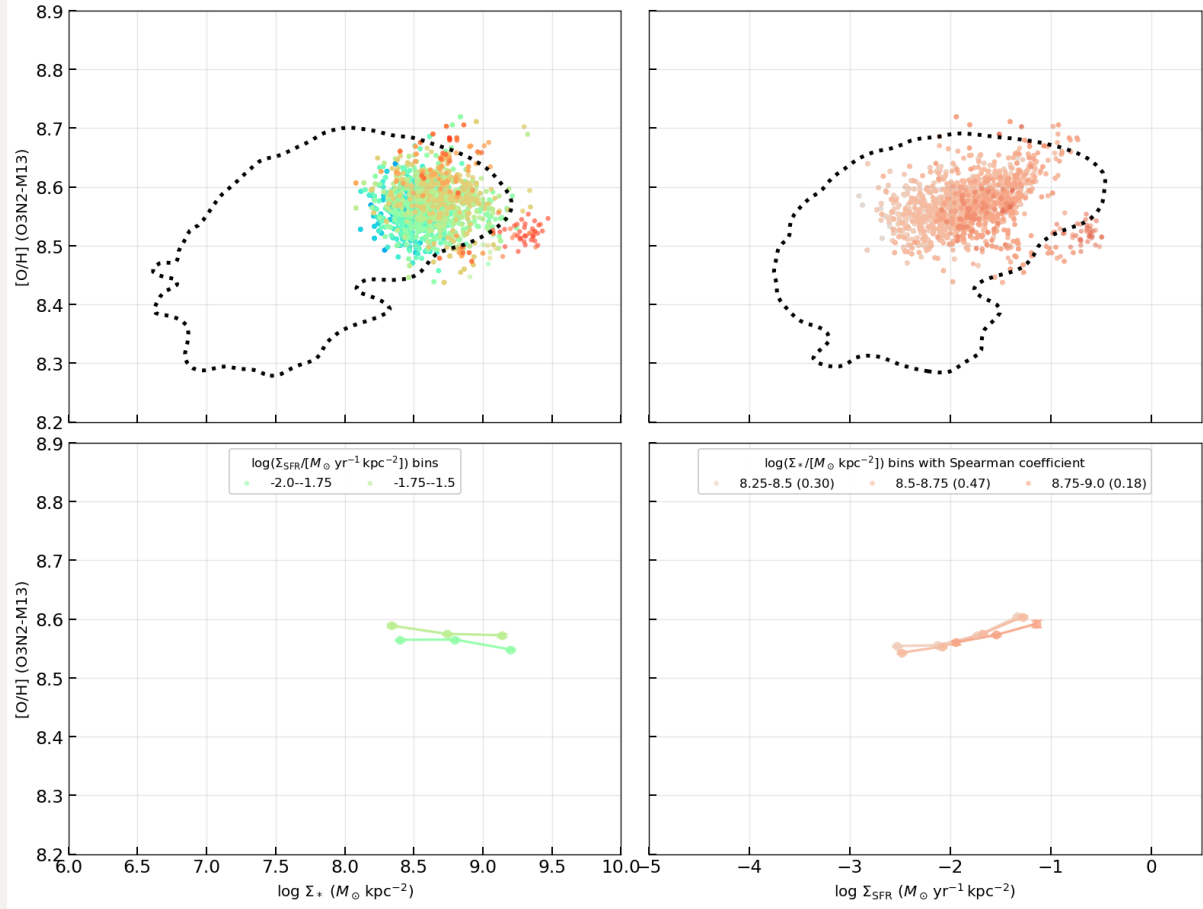
NGC4402: rMZR and Z_gas-Sigma_SFR relation (further_ready)



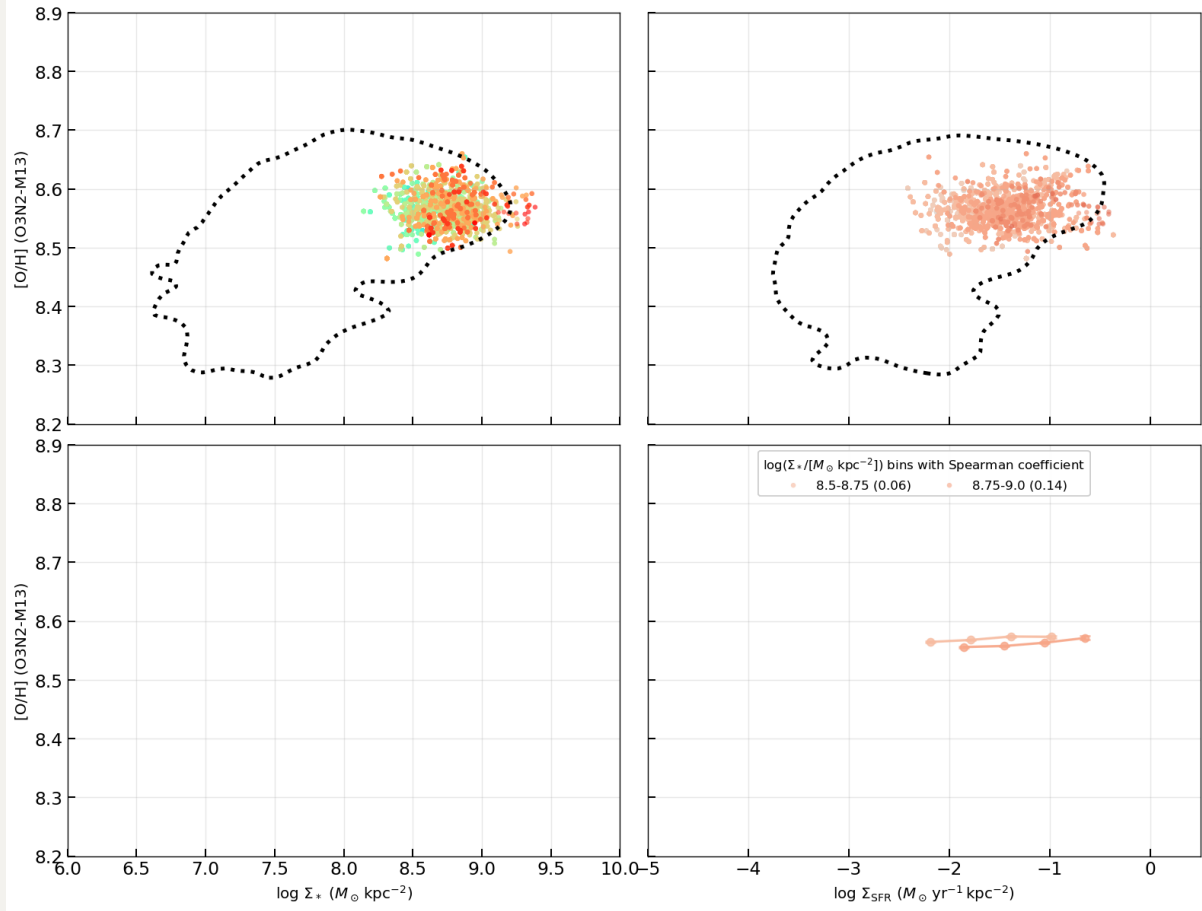
NGC4405: rMZR and Z_gas-Sigma_SFR relation (further_ready)



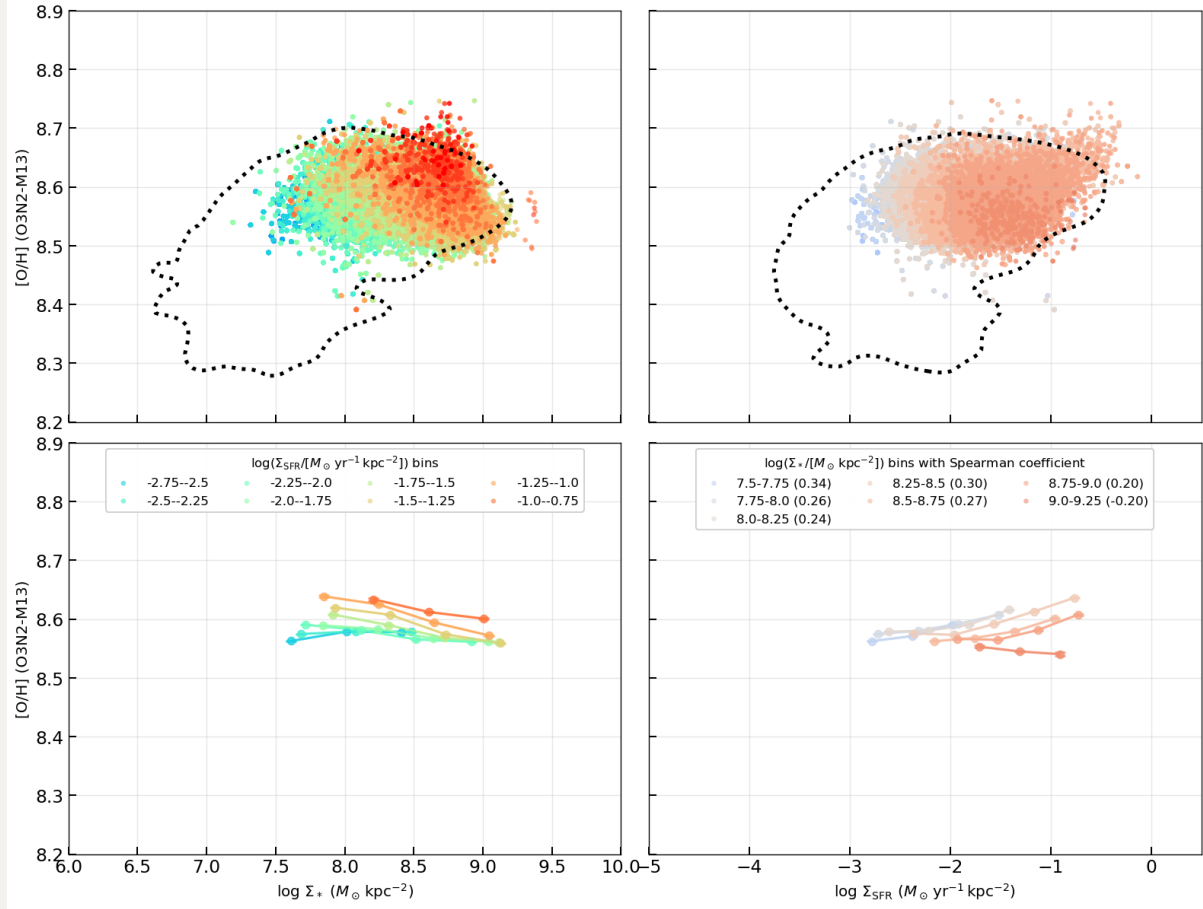
NGC4419: rMZR and Z_gas-Sigma_SFR relation (extended_fake)



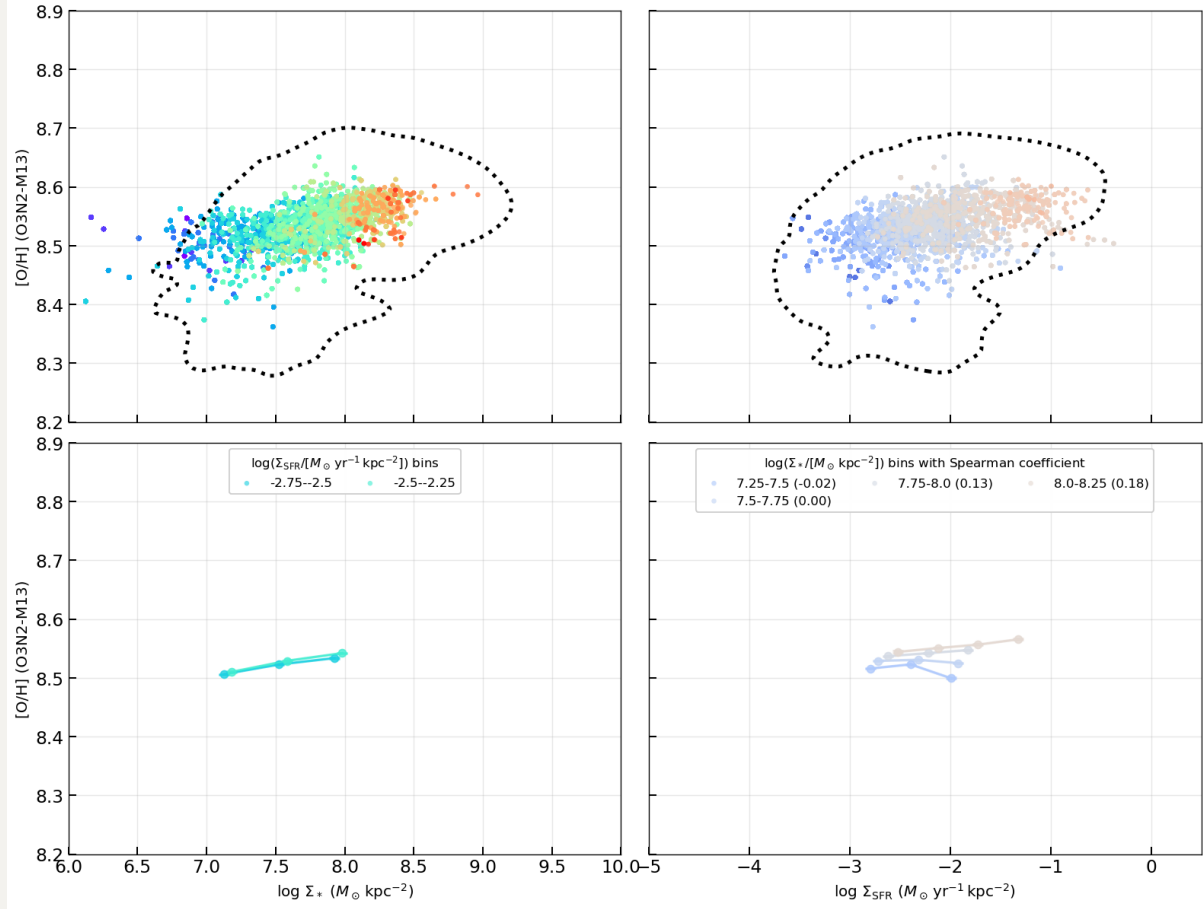
NGC4457: rMZR and Z_gas-Sigma_SFR relation (extended_fake)



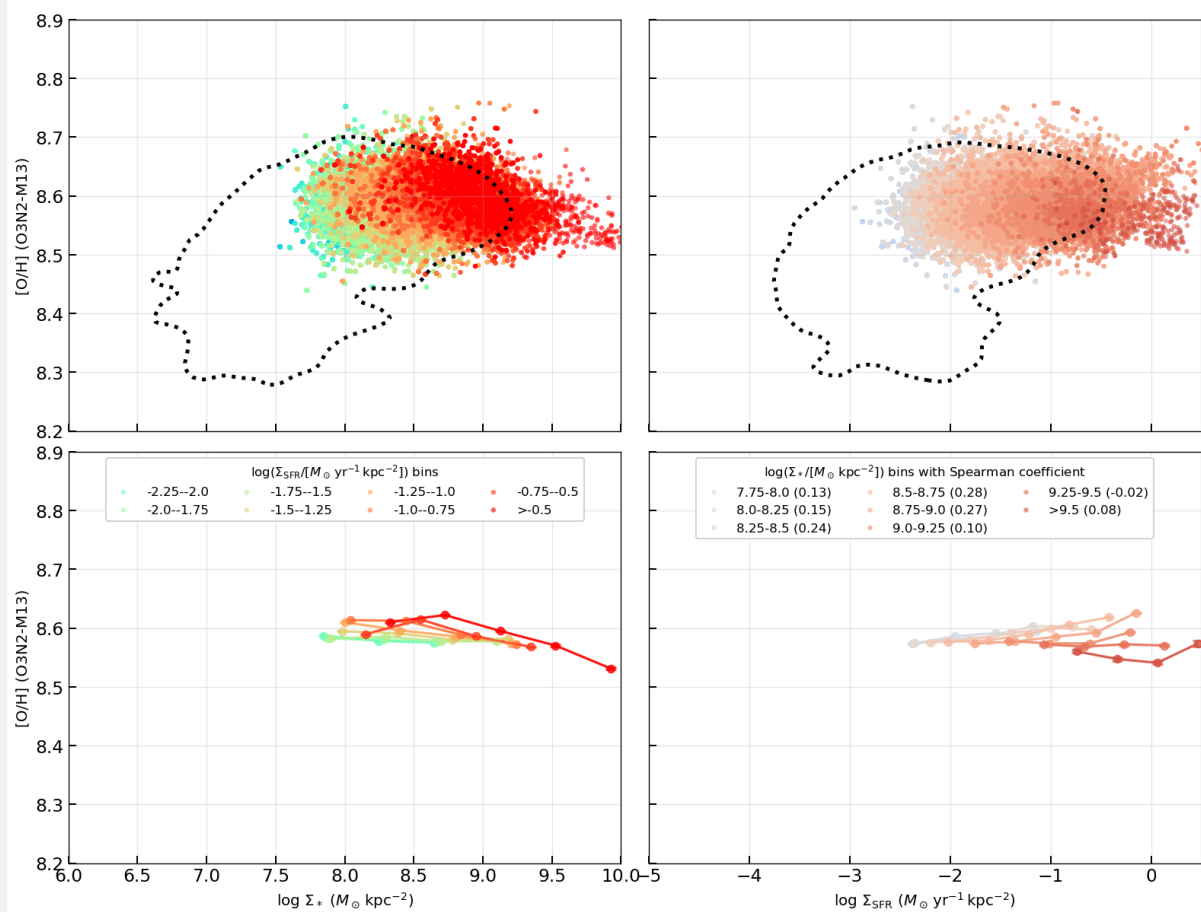
NGC4501: rMZR and Z_gas-Sigma_SFR relation (further_ready)



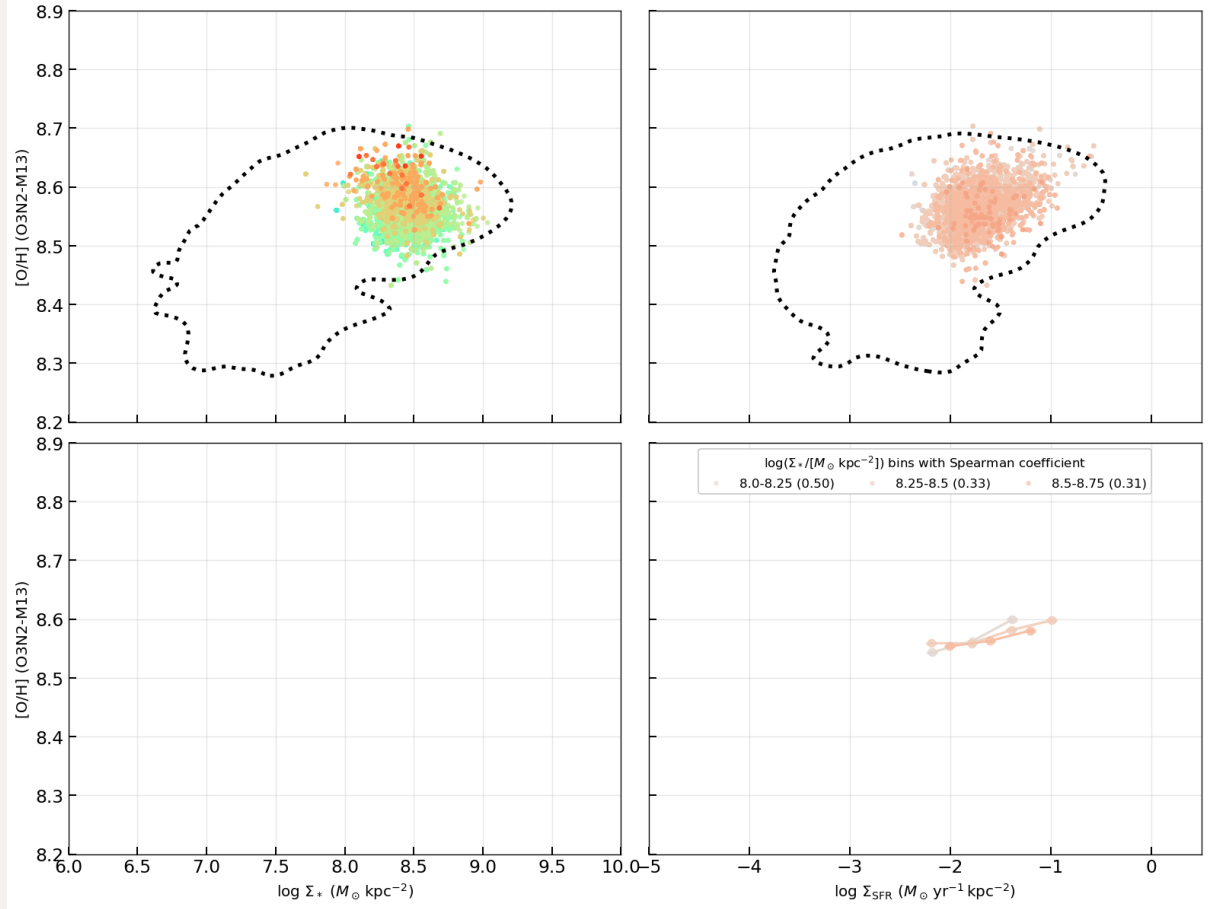
NGC4522: rMZR and Z_gas-Sigma_SFR relation (further_ready)



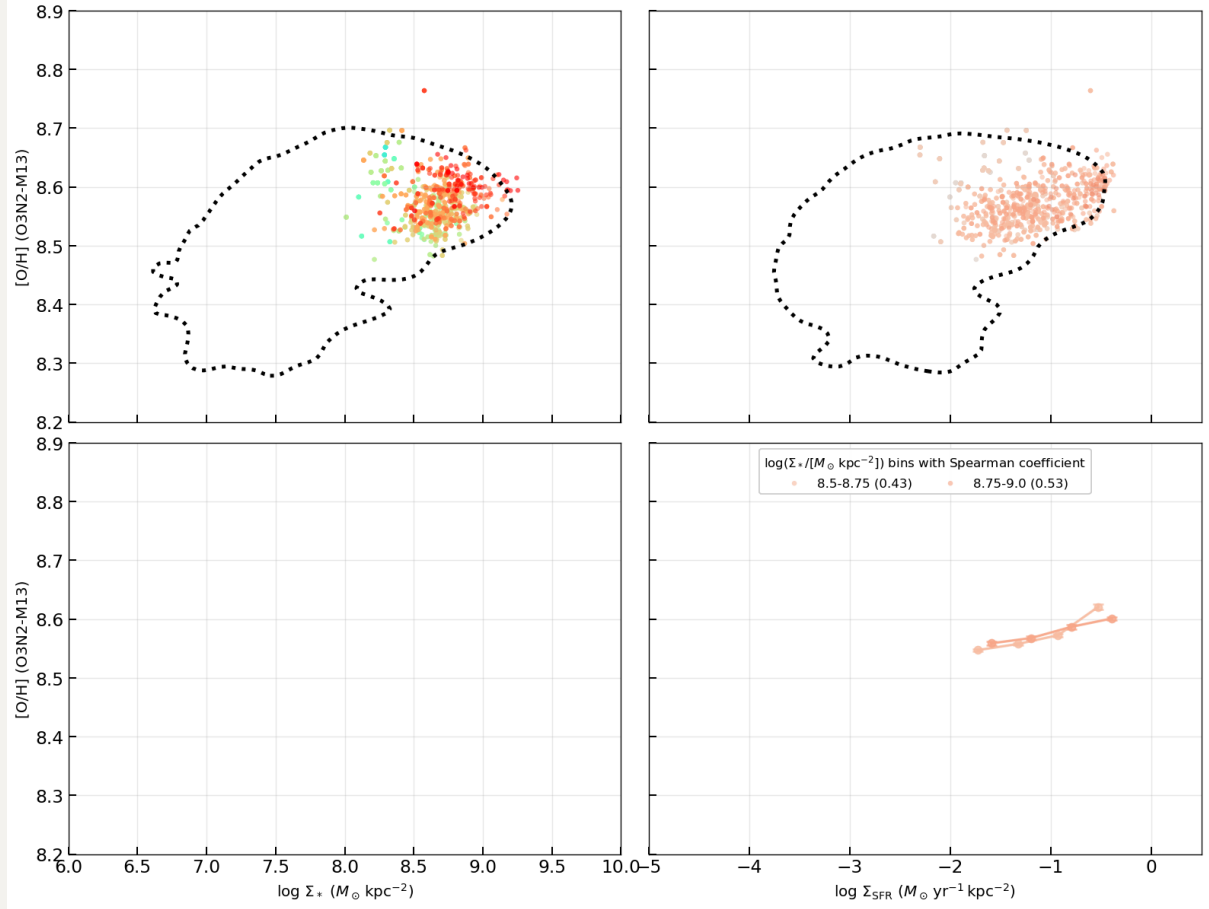
NGC4567_8: rMZR and Z_gas-Sigma_SFR relation (further_ready)



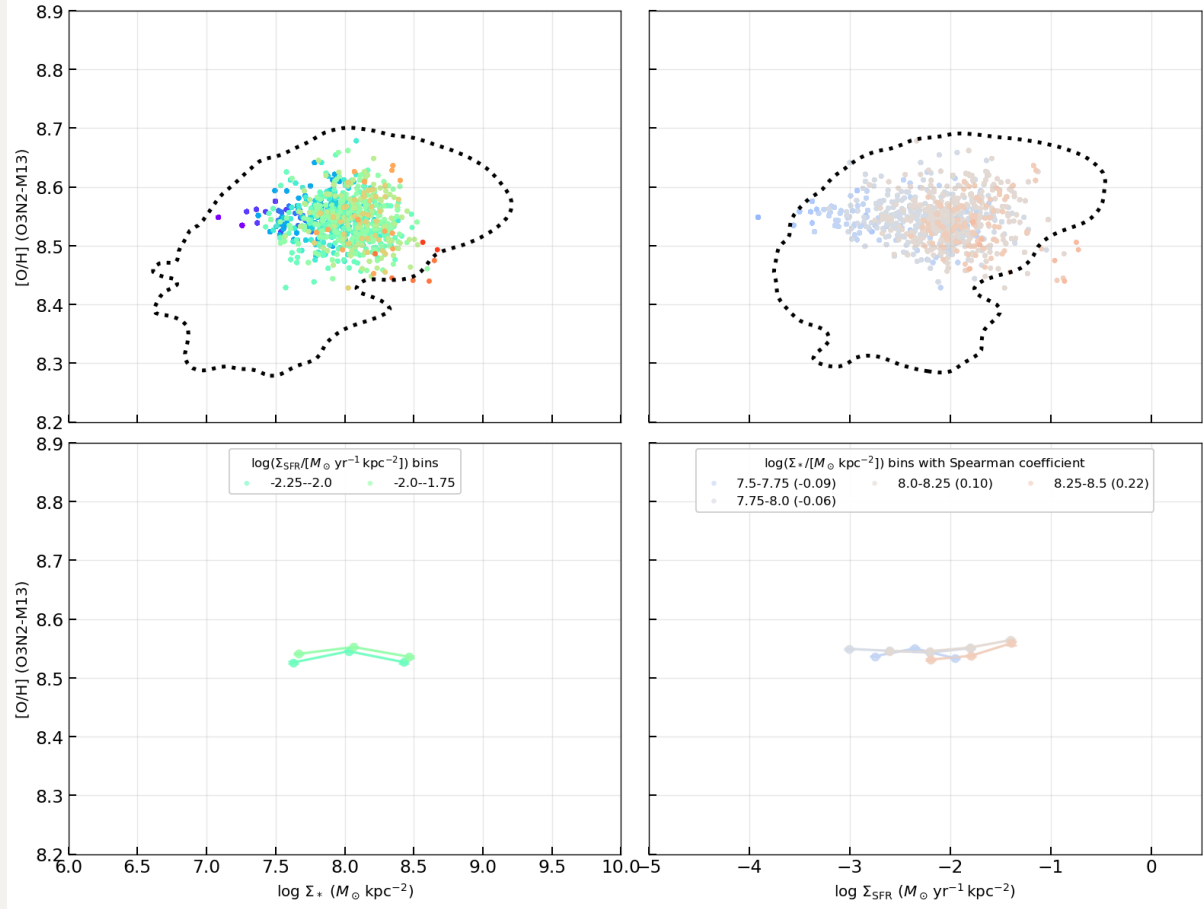
NGC4580: rMZR and Z_gas-Sigma_SFR relation (further_ready)



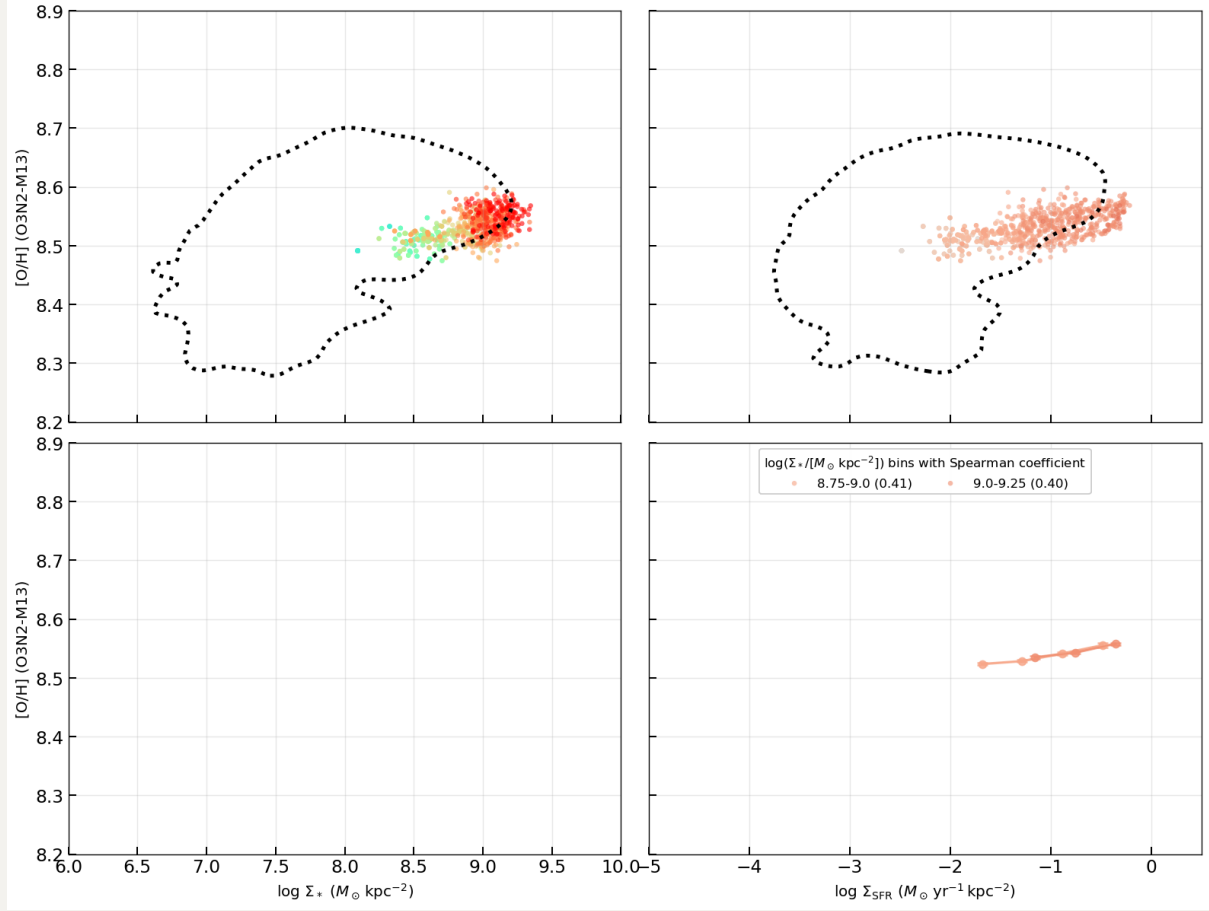
NGC4606: rMZR and Z_gas-Sigma_SFR relation (further_ready)



NGC4607: rMZR and Z_gas-Sigma_SFR relation (further_ready)



NGC4694: rMZR and Z_gas-Sigma_SFR relation (further_ready)



NGC4698: rMZR and Z_gas-Sigma_SFR relation (further_ready)

